ST13

Agriculture, Horticulture, Forestry, Animal Husbandry (Rajasthan focus); S&T Policy and Indian Scientists

Worth 3-5 questions in a typical paper. The hardest-working repeat here is the research-institution cluster - National Research Centre on Seed Spices at Ajmer, NRC on Camel at Bikaner, Central Institute for Arid Horticulture at Bikaner (2016, 2018, 2023) and the Udaipur Solar Observatory (2018 and 2021 - the identical question). Crop leadership, breeds by home tract and scientist-to-contribution matching fill the rest.

Two chapters bolted together, and both are Rajasthan-heavy. The first half is the applied science of a desert state - what grows in sand, how you save water, which breed belongs to which tract. The second half is institutional: the research centres sitting in Rajasthan, and the scientists whose names RPSC matches to their contributions. Almost none of it needs reasoning; all of it needs clean, district-level memory.

Agriculture, Livestock & S&T

Soil & nutrients

- 17 essential nutrients
- N chlorosis, P purple, K scorch
- Zn - khaira of rice
- Soil Health Card 19 Feb 2015, Suratgarh
- Neem-coated urea 2015

Seeds

- Breeder - golden yellow
- Foundation - white
- Certified - azure blue
- Truthfully labelled - opal green
- HHB-67 Improved (bajra)

Rajasthan crops

- No.1 in bajra, mustard, guar
- No.1 in coriander, cumin, methi
- Isabgol, henna, moth bean
- Guar is KHARIF
- Dryland below 750 mm

Water

- PMKSY 2015
- Har Khet Ko Pani
- Per Drop More Crop
- Drip ~90%, sprinkler 70-80%
- Khadin, tanka, nadi, johad

Horticulture

- Kinnow - Sri Ganganagar
- Orange - Jhalawar
- Olive - Lunkaransar, Bassi
- Date palm - Jaisalmer, Bikaner
- Sojat henna GI

Livestock

- Rathi best milch
- Chokla finest wool
- Magra carpet wool
- Sirohi goat, Jakhrana milk
- Camel state animal 2014

Institutions

- CAZRI Jodhpur 1959
- CSWRI Avikanagar 1962
- NRC Camel Bikaner 1984
- CIAH Bikaner 1993
- NRC Seed Spices Ajmer 2000
- Udaipur Solar Observatory 1975

S&T policy

- CSIR 1942, DST 1971
- STIP 2020 - fifth policy
- ANRF Act 2023
- NSD 28 Feb, NTD 11 May
- Space Day 23 Aug, Maths 22 Dec

Soil fertility and the plant nutrients

A plant needs seventeen essential nutrients. Three come free from air and water - carbon, hydrogen, oxygen. The rest must come from the soil, and they split into two classes by QUANTITY needed, not by importance. Primary macronutrients are nitrogen, phosphorus and potassium; secondary macronutrients are calcium, magnesium and sulphur; everything else is a micronutrient, needed in traces. RPSC's standard question is to slip calcium into a list of micronutrients and see if you notice.

- Primary macronutrients - nitrogen (N), phosphorus (P), potassium (K). The NPK you buy in a fertiliser bag.
- Secondary macronutrients - calcium, magnesium, sulphur.
- Micronutrients - iron, manganese, zinc, copper, boron, molybdenum, chlorine, nickel. Eight of them.
- Nitrogen builds protein and chlorophyll; phosphorus builds roots and grain; potassium regulates water and disease resistance.
- Zinc deficiency is widespread in the calcareous and sandy soils of Rajasthan - remember this as a state-specific fact.

Deficiency symptoms - a ready-made matching question

Nutrient	Classic symptom
Nitrogen	General chlorosis (yellowing) of the OLDER leaves
Phosphorus	Purplish tinge on leaves and stems
Potassium	Scorching or burning of the leaf margins
Zinc	Khaira disease of rice
Iron	Chlorosis of the YOUNG leaves, between the veins

- Soil Health Card Scheme - launched by the Prime Minister on 19 February 2015 from Suratgarh in Sri Ganganagar district, Rajasthan. A Rajasthan launch site, so learn the place, not just the year.
- Neem-coated urea was made mandatory in 2015 to slow nitrogen release and improve use efficiency.
- Manure adds bulk organic matter and improves soil structure; fertiliser adds concentrated nutrients and nothing else.
- Integrated Nutrient Management combines fertiliser, organic manure and biofertiliser so that soil health is not mined away.
- Rajasthan's soils - desert and arid sandy soils in the west, alluvial in the east, medium black in the Hadoti plateau, and saline-alkaline patches around the salt lakes.

TRAP

Calcium is NOT a micronutrient. Calcium, magnesium and sulphur are SECONDARY MACROnutrients. RPSC asked exactly this as a 'which one is not a micronutrient' question, with calcium as the answer. Second trap in the same area: nitrogen chlorosis shows on the OLDER leaves, iron chlorosis on the YOUNGER ones.

Seeds - the classes and the multiplication chain

Seed is not one thing; it moves down a certified ladder, and each rung has its own tag colour. The breeder produces a tiny quantity of breeder seed under his own supervision. That is multiplied into foundation seed, and foundation seed into certified seed, which is what actually reaches the farmer. The order and the colours are both asked.

Seed multiplication chain - order and tag colour

Nucleus seed (with the breeder) → Breeder seed - GOLDEN YELLOW tag → Foundation seed - WHITE tag → Certified seed - AZURE BLUE tag → Farmer's field

- Truthfully labelled seed carries an OPAL GREEN label; it is labelled by the producer, not certified by an agency.
- HYV = high-yielding variety, a pure line that can be saved and resown. Hybrid = the F1 cross of two parents; its yield collapses in the next generation, so seed must be bought afresh every year.
- That single difference - hybrid seed cannot be saved - is why hybrid adoption depends on a working seed supply chain.
- Genetically modified seed carries an inserted gene; in India only Bt cotton is approved for commercial cultivation.
- Seed replacement rate measures how much of the area is sown with fresh certified seed each season.

YAAD RAKHO

Fix the ladder as B-F-C - Breeder, Foundation, Certified - and hang the colours on it in that order: GOLDEN yellow, WHITE, azure BLUE. Say it as 'gold, white, blue' going down the ladder. Opal green sits outside the ladder altogether, on truthfully labelled seed.

Rajasthan's crops and the varieties bred for them

Rajasthan's agriculture is defined by scarcity, so the state leads India in exactly the crops that tolerate it - coarse millets, oilseeds, gums and seed spices. Learn the leadership list as a block, because RPSC asks it as a statement question with one wrong crop slipped in (sugarcane is the usual planted error - that is Uttar Pradesh).

- Rajasthan is India's LARGEST producer of: bajra, rapeseed-mustard, guar, coriander, cumin, fenugreek (methi), isabgol, henna and moth bean.
- It is also India's largest wool producer and holds the country's largest camel population.
- Kharif crops of Rajasthan - bajra, jowar, maize, moth, guar, cotton, groundnut, soybean, til.
- Rabi crops - wheat, barley, gram, mustard, taramira, isabgol. Barley is RABI; it has been used as the odd one out in a kharif list.
- Guar is a KHARIF crop, sown with the onset of the monsoon; its gum is exported for use in the oil and gas industry. Main districts - Sri Ganganagar, Hanumangarh, Bikaner, Churu, Nagaur.
- The ICAR Directorate of Rapeseed-Mustard Research is at Sewar, Bharatpur - the national mustard institute sits in Rajasthan.
- Crop rotation with a legume restores fertility because root-nodule bacteria fix atmospheric nitrogen. Mixed cropping sows two crops together as insurance; intercropping sows them in a defined row pattern.

Varieties and hybrids grown in Rajasthan - learn crop against name

Crop	Varieties / hybrids	Why it matters here
Bajra (pearl millet)	HHB-67 Improved is the flagship; the HHB and RHB hybrid series dominate the state	HHB-67 Improved is early-maturing and downy-mildew resistant - it escapes drought by finishing fast
Maize	Mahi Kanchan and Mahi Dhawal (Banswara centre); the Pratap Makka series from MPUAT Udaipur	Bred for the southern Rajasthan maize belt - Banswara, Dungarpur, Udaipur, Chittorgarh

Crop	Varieties / hybrids	Why it matters here
Mustard (rapeseed-mustard)	Pusa Bold and Bio-902	Rajasthan is the country's largest producer; the national institute is at Bharatpur
Guar (cluster bean)	RGC-936 and RGC-1003	Bred for the north-western guar tract; guar gum is a major export earner
Wheat	The 'Raj' series of varieties bred in the state	Rabi crop of the canal-irrigated north and the eastern plain; not a Rajasthan leadership crop

TRAP

Three regular traps. One: guar is KHARIF, not rabi. Two: Rajasthan does NOT lead in sugarcane - that is Uttar Pradesh, and it is the wrong statement RPSC plants in the leadership list. Three: HHB-67 Improved is a BAJRA hybrid, not a wheat or maize one; it is offered against all three.

Dryland farming, water-use efficiency, drip and sprinkler

Dryland farming is agriculture in areas that get less than about 750 mm of rain a year, without assured irrigation. Its whole objective is not raising yield but CONSERVING SOIL MOISTURE - every technique follows from that one aim. Most of western Rajasthan falls in this category, which is why khadin cultivation, mulching and short-duration drought-escaping varieties are standard here rather than exotic.

- Dryland farming - under about 750 mm annual rainfall; rainfed farming can extend above that where rain is more reliable.
- Moisture conservation techniques - mulching (cuts evaporation from the soil surface), contour bunding, deep ploughing before the monsoon, weed control, and wind-breaks.
- Khadin, of the Jaisalmer region, is a runoff-harvesting cropping system: a bund across a slope impounds runoff so a rabi crop can be grown on stored moisture.
- Other traditional structures - tanka, nadi, johad, baori and toba. These recur across the Rajasthan Geography paper too.
- Water-use efficiency, roughly: surface or flood irrigation 30-50 per cent, sprinkler 70-80 per cent, drip about 90 per cent.
- Pradhan Mantri Krishi Sinchayee Yojana (2015) carries two slogans you must attach correctly - 'Har Khet Ko Pani' for coverage and 'Per Drop More Crop' for micro-irrigation.

Drip vs sprinkler - which one for which field

Drip irrigation	Sprinkler irrigation
Water delivered drop by drop at the root zone through emitters	Water sprayed over the crop like rainfall through nozzles
Efficiency about 90 per cent - the highest of all methods	Efficiency about 70-80 per cent
Best for wide-spaced row crops and orchards - kinnow, pomegranate, date palm, vegetables	Best for close-spaced crops on the sandy soils of western Rajasthan - wheat, groundnut, mustard, fodder
Allows fertigation; keeps the leaf dry, so less fungal disease	Cuts conveyance and deep-percolation losses that ruin flood irrigation on sand

- Protected cultivation structures - greenhouse, poly-house, shade-net house, walk-in tunnel. Khadin is NOT one of them; it is a water-harvesting field system, and that is a standard trick option.
- Precision farming uses soil maps, sensors and variable-rate application to match input to need.
- Namo Drone Didi supplies agricultural drones to women's self-help groups for spraying and monitoring.
- Post-harvest technology - grading, cold storage, controlled-atmosphere storage; post-harvest loss is the silent tax on horticulture.

CURRENT LINK

CURRENT LINK, as of September 2026 - the Rajasthan Budget 2026-27 was presented on 11 February 2026 by Deputy Chief Minister Diya Kumari, who also holds Finance. Its farm announcements ran along familiar lines: PM-KUSUM solar pumps with a target of about 50,000 installations, micro-irrigation coverage of roughly three lakh hectares, farm electricity and interest subsidy, crop and livestock insurance, and gaushala support, framed within the 'Viksit Rajasthan @2047' vision. Rajasthan has presented a separate agriculture budget since 2022-23 - that practice itself is an examinable fact.

Pests, the locust threat and integrated pest management

Every arid-zone syllabus in India carries the locust because western Rajasthan is on its invasion path. The species is the desert locust, Schistocerca gregaria. Its danger lies in a phase change: in the solitary phase it is a harmless grasshopper, but when rain and vegetation let numbers build, it switches to the gregarious phase, changes colour, and moves in swarms that strip a field bare in hours. India's answer is a permanent watch, and the headquarters of that watch is in Rajasthan.

How a locust warning travels - the chain to learn

FAO Desert Locust Information Service issues global bulletins →

Locust Warning Organisation, Jodhpur - the national headquarters →

Circle offices survey the Scheduled Desert Area in Rajasthan and Gujarat →

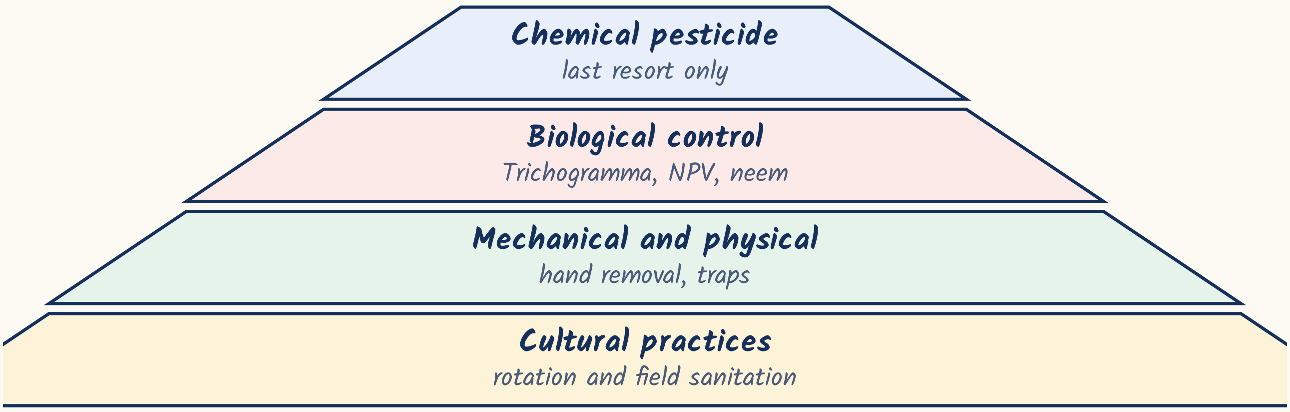
Control operations - spraying, now with vehicle-mounted and drone sprayers →

Reporting up to the Directorate of Plant Protection, Quarantine and Storage, Faridabad

- Desert locust = Schistocerca gregaria. Solitary phase harmless, gregarious phase destructive.
- The Locust Warning Organisation is headquartered at JODHPUR, under the Directorate of Plant Protection, Quarantine and Storage at Faridabad.
- The Scheduled Desert Area for locust control lies in Rajasthan and Gujarat.
- The last major upsurge reached Rajasthan in 2019-20; drones were deployed for control spraying in 2020.
- Other important pests of Rajasthan crops - white grub in groundnut, termite in the sandy tract, pod borer in gram, and downy mildew in bajra.

- Integrated Pest Management puts cultural, mechanical and biological methods FIRST and treats chemical pesticide as the last resort.
- Its objective is to hold the pest below the Economic Threshold Level - NOT to eradicate the species. That distinction is the standard wrong statement.
- Trichogramma is an egg parasitoid used against lepidopteran (caterpillar) pests; NPV and neem-based formulations are other biocontrol tools.
- Biofertilisers - Rhizobium, Azotobacter, Azospirillum, blue-green algae, mycorrhiza. Biopesticides - Bacillus thuringiensis, Trichoderma.
- Pesticide residue and resistance are the two costs of a chemical-first approach; IPM exists to avoid both.

Integrated Pest Management - chemical is the last resort



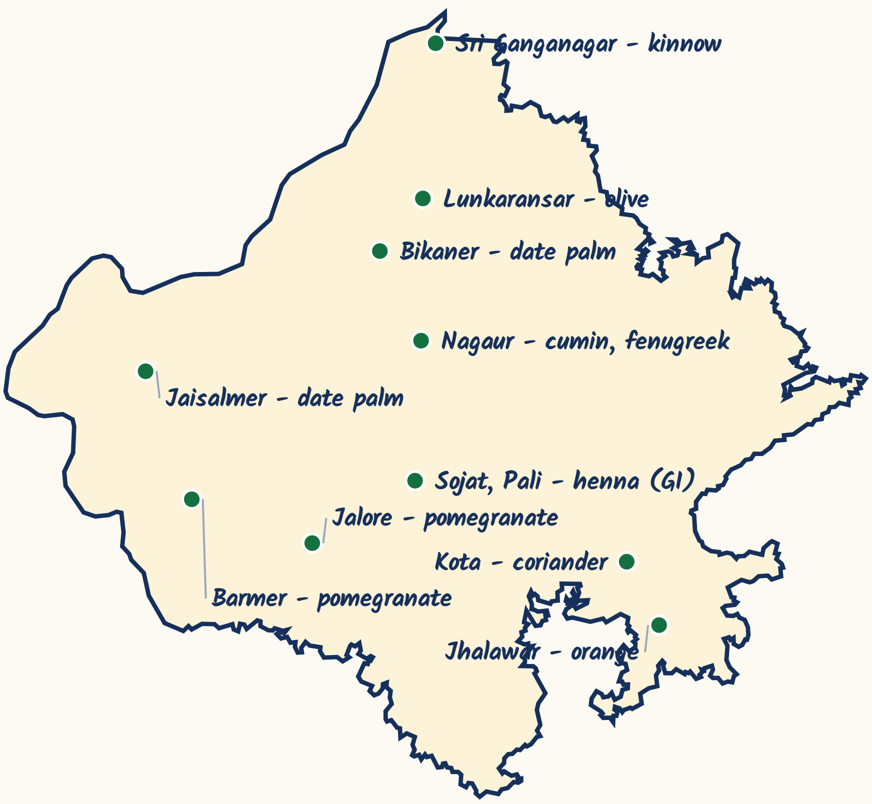
CURRENT LINK

CURRENT LINK, as of September 2026 - no major locust upsurge has reached Rajasthan since the 2019-20 invasion. The Locust Warning Organisation at Jodhpur continues routine survey of the Scheduled Desert Area and FAO continues to publish desert locust bulletins for the region. If a swarm year does occur before the exam, the examinable facts stay the same: Schistocerca gregaria, LWO Jodhpur, Rajasthan and Gujarat.

Horticulture – fruit belts, the seed spice bowl and medicinal plants

Horticulture in Rajasthan means arid horticulture – fruit that survives heat, sand and salinity. Learn each fruit against its belt, because this is asked as a 'which pair is NOT correctly matched' question. The classic wrong pair offered is olive with Jhalawar; Jhalawar is oranges, and olive is Lunkaransar and Bassi.

Fruit and seed-spice belts of Rajasthan



Olive is Lunkaransar and Bassi – Jhalawar is orange

Fruit and its growing belt in Rajasthan

Crop	Belt
Kinnow	Sri Ganganagar and Hanumangarh
Orange	Jhalawar

Crop	Belt
Olive	Lunkaransar (Bikaner) and Bassi (Jaipur) - the state's olive plantation and refinery project
Date palm	Jaisalmer and Bikaner, through tissue-culture plantations
Pomegranate	Barmer and Jalore
Ber and aonla	The arid tract generally - the core mandate crops of the Bikaner institute

- Seed spices of Rajasthan - coriander, cumin, fennel, fenugreek, ajwain, dill, celery and nigella. Rajasthan and Gujarat together dominate India's seed-spice output.
- Coriander belongs mainly to the Kota-Baran-Jhalawar-Chittorgarh belt; cumin to the western districts around Jodhpur, Barmer, Jalore and Nagaur; fenugreek to Nagaur and Sikar.
- Rajasthan is called India's seed spice bowl, and its dedicated institute is the National Research Centre on Seed Spices at Tabiji, Ajmer.
- Sojat in Pali district is the henna capital of India and its henna carries a Geographical Indication tag.
- Ashwagandha grown around Nagaur is known locally as asgandh; guggul is a threatened resin-yielding shrub of the Aravalli and desert tracts.
- Floriculture and vegetable cultivation cluster around the peri-urban belts of Jaipur, Ajmer and Kota; the national horticulture programme is MIDH.

Medicinal and aromatic plants - botanical names get matched

Common name	Botanical name
Henna (mehndi)	Lawsonia inermis
Isabgol	Plantago ovata
Ashwagandha	Withania somnifera
Safed musli	Chlorophytum borivilianum
Guggul	Commiphora wightii

ASKED IN RAS

Asked in RAS Pre 2016 - the Central Institute for Arid Horticulture at Bikaner works on arid fruits such as ber, date palm, aonla and pomegranate, and was established in 1993 as the National Research Centre for Arid Horticulture. The statement RPSC marks WRONG is that it works under the Ministry of Environment - like every ICAR institute it sits under the Department of Agricultural Research and Education in the Ministry of Agriculture and Farmers Welfare.

Forestry and agroforestry - the khejri system

Agroforestry in Rajasthan has one central tree, and its name is khejri. Prosopis cineraria is the state tree and is called the kalpavriksha of the desert because it does everything at once - it fixes nitrogen, its light canopy lets the crop below get sun, its leaves (loong) feed livestock, and its pods, sangri, are a food. A bajra or moth field with khejri standing in it is the classic agri-silviculture system of the desert, and it is the reason the Bishnoi tradition of protecting this tree - Khejarli, 1730, Amrita Devi - runs so deep.

- Khejri (Prosopis cineraria) - state tree; pods called sangri; nitrogen-fixing; the backbone of desert agroforestry.
- Rohida (Tecomella undulata) - state flower, called Marwar teak or the teak of the desert.
- Prosopis juliflora is the INVASIVE 'vilayati babool'. One word apart from khejri (Prosopis CINERARIA) and the opposite answer - never let the two merge.
- Agroforestry systems - agri-silviculture (crops plus trees), silvipasture (grass plus trees for livestock), horti-pasture.
- India was the first country in the world to adopt a dedicated National Agroforestry Policy, in 2014.
- Oran and gochar are the community sacred groves and pastures of Rajasthan - in-situ conservation without any statute behind it.

- National Forest Policy 1988 - target of 33 per cent of the geographical area under forest and tree cover, 60 per cent in hills and 20 per cent in the plains.
- That policy shifted the object of forestry from REVENUE to ecological stability and the needs of forest-dependent communities. 'Maximising revenue' is the wrong statement RPSC uses.
- Joint Forest Management was formally introduced by a Government of India circular in 1990 - village committees share protection duties and usufruct.
- Van Dhan Vikas Kendras add value to minor forest produce and are run by TRIFED under the Ministry of Tribal Affairs.
- Non-timber forest produce of Rajasthan - tendu leaf, gum from khair and kumat, honey, sangri, aonla and medicinal roots.

Animal husbandry - breeds by tract, and the diseases

Breed questions are pure matching, and the trick is to learn the HOME TRACT, not the animal's looks. Fix one anchor fact per breed - Rathi is the best milch cow, Chokla gives the finest wool, Magra gives lustrous carpet wool, Sirohi is the meat goat, Jakhrana of Alwar is the milk goat. With those anchors the tables fall into place.

Home tracts of Rajasthan's livestock breeds

Learn the tract, not the animal - this is asked as a four-pair match

Cattle breeds of Rajasthan and their home tracts

Breed	Home tract	Anchor fact
Rathi	Bikaner and Sri Ganganagar	The best MILCH breed of the state
Tharparkar	Jaisalmer and Barmer	Dual purpose - milk and draught
Kankrej	Jalore and Sirohi	Prized draught breed, also good milk
Nagauri	Nagaur	Pure draught breed
Malvi	Jhalawar	Draught, from the Malwa side
Hariana	Bharatpur and Alwar	Comes in from the Haryana tract

Sheep breeds - wool is the anchor

Breed	Home tract	Anchor fact
Chokla	Sikar and Jhunjhunu (Shekhawati)	FINEST wool among Indian sheep breeds - the 'jewel of Rajasthan'
Magra	Bikaner and Jaisalmer	Lustrous wool, prized for carpets
Marwari	Jodhpur and Pali	Hardy, coarse wool, widely kept
Nali	Sri Ganganagar and Churu	Carpet wool
Sonadi	Udaipur and Dungarpur	Southern tract, also gives milk
Pugal, Jaisalmeri, Malpuri	Western and eastern tracts	Names to recognise in an option set

- Goat breeds - Sirohi (meat), Jakhrana of ALWAR (milk), Marwari, Barbari.
- Camel breeds - Bikaneri, Jaisalmeri (the riding breed), Mewari (the milk breed), Kachchhi.
- The camel was declared Rajasthan's state animal (domestic) in 2014 and protected by a state Act of 2015; the chinkara is the state animal (wild). Rajasthan has India's largest camel population.
- Buffalo breeds seen in the state - Murrah, Surti, Badhawari, Jafarabadi. Horse - Marwari and Kathiawari.
- Breeding technology - artificial insemination and embryo transfer are the standard routes for spreading superior germplasm.
- Dairy in Rajasthan runs through RCDF and the Saras brand on the Amul co-operative model; RAJUVAS at Bikaner is the state veterinary university.
- The state also runs livestock breeding farms - the Chandan sheep breeding farm in Jaisalmer is the best known - and supports gaushalas and Nandi shalas through departmental schemes.

- Foot-and-mouth disease - VIRAL, affects cloven-hoofed animals. PPR (peste des petits ruminants) - VIRAL, of sheep and goat.
- Brucellosis - BACTERIAL and zoonotic, caused by Brucella. This is the odd one out in a 'which is not viral' question.
- Lumpy skin disease - caused by a capripoxvirus, spread mainly by biting insects, affects cattle and buffalo. It does NOT infect humans through milk; that is the standard wrong statement.
- The 2022 lumpy skin outbreak caused heavy cattle mortality in Rajasthan; the indigenous Lumpi-ProVacInd vaccine followed.
- National Animal Disease Control Programme (2019) targets foot-and-mouth disease and brucellosis specifically.
- Rashtriya Gokul Mission (2014) develops and conserves indigenous bovine breeds; the National Livestock Mission covers feed, fodder and small ruminants.

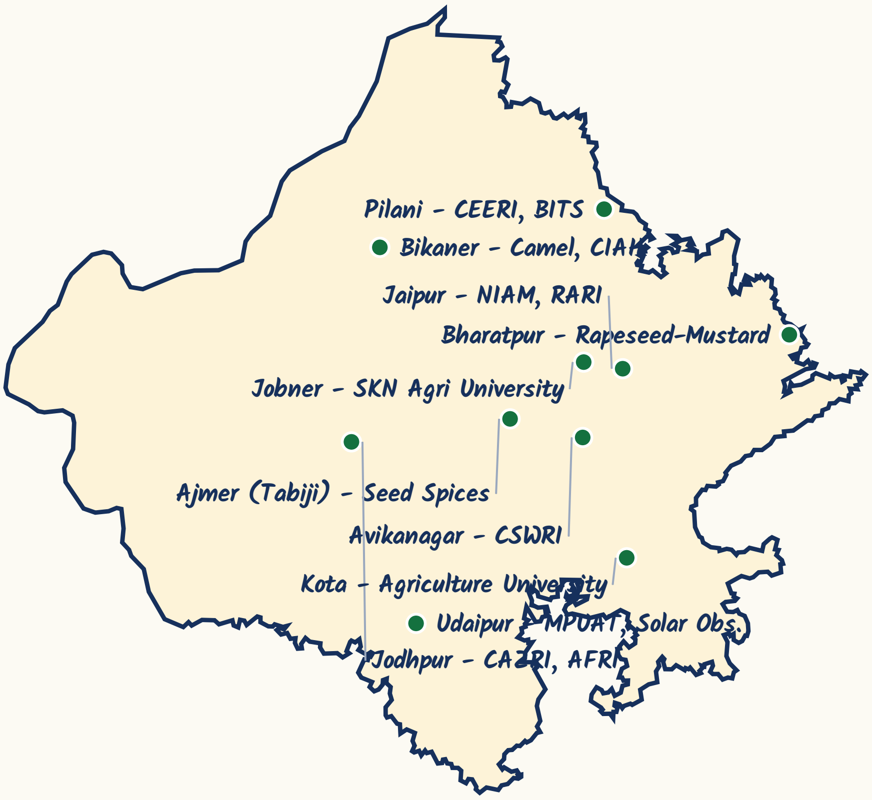
ASKED IN RAS

Asked in RAS Pre 2018 - where is the National Research Centre on Camel located? Answer: Bikaner, established in 1984. The Central Institute for Arid Horticulture is also at Bikaner, so the city carries two ICAR institutes and RPSC uses that overlap in matching questions.

The research institutions of Rajasthan - the most repeated block in this chapter

Read this section twice. It is the densest repeat zone in the whole of Science and Technology for Rajasthan - the institution cluster has been asked in 2016, 2018, 2021 and 2023, sometimes as a straight location question, sometimes as a four-pair match, sometimes as a 'which one is NOT in Rajasthan'. Learn the institute, the town, the district where the town is not obvious, and the year.

Research institutions of Rajasthan



Bikaner and Jodhpur each carry more than one - that is the trick

Research institutions located in Rajasthan

Institution	Place	Year / note
ICAR-Central Arid Zone Research Institute (CAZRI)	Jodhpur	1959; India's premier arid-zone research institute
ICAR-Central Sheep and Wool Research Institute (CSWRI)	Avikanagar, Malpura tehsil, TONK district	1962
ICAR-National Research Centre on Camel	Bikaner	1984

Institution	Place	Year / note
ICAR-Central Institute for Arid Horticulture (CIAH)	Bikaner	1993, as the NRC for Arid Horticulture
ICAR-National Research Centre on Seed Spices	Tabiji, Ajmer	2000
ICAR-Directorate of Rapeseed-Mustard Research	Sewar, Bharatpur	The national mustard institute
Arid Forest Research Institute (AFRI)	Jodhpur	Under ICFRE, not ICAR
CSIR-Central Electronics Engineering Research Institute (CEERI)	Pilani	The ONLY CSIR laboratory in Rajasthan
Udaipur Solar Observatory	An island in Fateh Sagar Lake, Udaipur	1975, founded by Dr. Arvind Bhatnagar; run by PRL Ahmedabad; hosts MAST
Rajasthan Remote Sensing Centre	Jodhpur	State remote sensing agency
Ch. Charan Singh National Institute of Agricultural Marketing (NIAM)	Jaipur	Agricultural marketing training
Rajasthan Agricultural Research Institute (RARI)	Durgapura, Jaipur	State research institute

The institutes in order of establishment – RPSC has asked this as a chronology

CAZRI Jodhpur 1959 → CSWRI Avikanagar 1962 → Udaipur Solar Observatory 1975 → NRC on Camel, Bikaner 1984 → CIAH Bikaner 1993 → NRC on Seed Spices, Ajmer 2000

ASKED IN RAS

Asked in RAS Pre 2023 - where is the ICAR National Research Centre on Seed Spices? Answer: Tabiji, Ajmer. And asked in RAS Pre 2018 and again in 2023 as a four-pair match - Seed Spices with Ajmer, Arid Horticulture with Bikaner, CAZRI with Jodhpur, CSWRI with Avikanagar (Tonk). This exact matching set is the highest-probability question in the chapter.

ASKED IN RAS

Asked in RAS Pre 2018 and again in RAS Pre 2021 - the identical question - the Udaipur Solar Observatory on the island in Fateh Sagar Lake is at present operated by which body? Answer: the Physical Research Laboratory, Ahmedabad. Twice in consecutive cycles; do not lose this mark. And in RAS Pre 2016 the negative version was asked - which institute is NOT in Rajasthan? Answer: the Central Institute for Research on Goats, which is at Makhdoom in Mathura district, Uttar Pradesh.

Universities of Rajasthan you must place

University	Headquarters
Maharana Pratap University of Agriculture and Technology (MPUAT)	Udaipur
Swami Keshwanand Rajasthan Agricultural University (SKRAU)	Bikaner
Sri Karan Narendra Agriculture University (SKNAU)	Jobner (Jaipur)
Agriculture University	Jodhpur, and a second at Kota
Rajasthan University of Veterinary and Animal Sciences (RAJUVAS)	Bikaner
Birla Institute of Technology and Science (BITS)	Pilani - the state's leading private deemed university

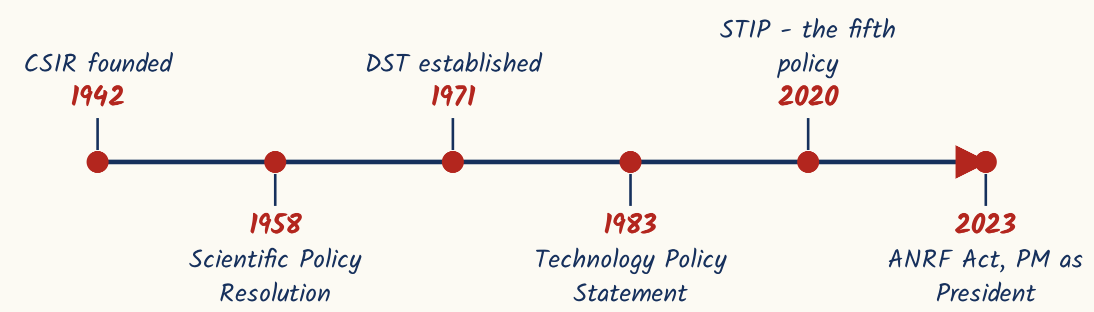
YAAD RAKHO

A city-wise trick. BIKANER holds two ICAR institutes - Camel and Arid Horticulture - plus SKRAU and RAJUVAS. JODHPUR holds three - CAZRI, AFRI and the Remote Sensing Centre - plus the Locust Warning Organisation. AJMER has Seed Spices, BHARATPUR has mustard, TONK has sheep and wool, PILANI has CEERI, UDAIPUR has the Solar Observatory and MPUAT. Learn the CITY first and the institutes hang off it.

S&T policy, the science days, and the scientists RPSC matches

The last block is institutional memory. Two dates matter most - CSIR was founded in 1942, before independence, and DST in 1971. On top of them sits a stack of five policy documents, of which the Science, Technology and Innovation Policy of 2020 is the fifth. Then learn the four national science days, and finally the scientist-to-contribution table, which is where the marks actually are.

India's science and technology architecture



- CSIR founded 1942; DST 1971; DSIR and DBT in the 1980s.
- India's five S&T policy documents - Scientific Policy Resolution 1958, Technology Policy Statement 1983, Science and Technology Policy 2003, STI Policy 2013, and STIP 2020 (the fifth).
- Anusandhan National Research Foundation - created by an Act of 2023, with the Prime Minister as ex-officio President of its Governing Board. It subsumed the Science and Engineering Research Board; it did NOT replace CSIR, which continues.
- Shanti Swarup Bhatnagar Prize - awarded by CSIR; INSPIRE stands for Innovation in Science Pursuit for Inspired Research, run by DST; Vigyan Jyoti encourages girls into STEM.
- The Physical Research Laboratory at Ahmedabad was founded in 1947 by Vikram Sarabhai - and it is PRL that runs the Udaipur Solar Observatory, which ties this section back to the last one.

The four national science days

Day	Date	What it commemorates
National Science Day	28 February	The Raman Effect, 1928
National Technology Day	11 May	The Pokhran-II tests, 1998
National Space Day	23 August	The Chandrayaan-3 landing, 2023
National Mathematics Day	22 December	Srinivasa Ramanujan's birth anniversary

Indian scientists matched to their contributions

Scientist	Contribution
C.V. Raman	Raman Effect - scattering of light; Nobel Prize in PHYSICS, 1930
Srinivasa Ramanujan	Number theory, partitions, infinite series; self-taught mathematician
Jagadish Chandra Bose	The crescograph, and work on plant response and radio waves
Meghnad Saha	The thermal ionisation equation
Satyendra Nath Bose	Bose-Einstein statistics; the boson is named after him

Scientist	Contribution
Homi J. Bhabha	Bhabha scattering; the three-stage nuclear programme; founded TIFR in 1945
Vikram Sarabhai	Founded PRL in 1947; father of India's space programme
Har Gobind Khorana	Genetic code and protein synthesis; Nobel in PHYSIOLOGY OR MEDICINE, 1968
Subrahmanyan Chandrasekhar	Chandrasekhar limit, stellar structure; Nobel in Physics, 1983
Venkatraman Ramakrishnan	Structure and function of the RIBOSOME; Nobel in Chemistry, 2009
M.S. Swaminathan	Father of the Green Revolution in India
Verghese Kurien	Operation Flood and the White Revolution; Amul and NDDB
Salim Ali	The 'Birdman of India'; his ornithological survey of Keoladeo Ghana, Bharatpur
Birbal Sahni	Palaeobotany; the institute at Lucknow carries his name

- Also learn: A.P.J. Abdul Kalam - the missile programme and Pokhran-II, later President of India.
- Daulat Singh Kothari - BORN AT UDAIPUR; chaired the Education Commission of 1964-66 and was the first Scientific Adviser to the Ministry of Defence. A Rajasthan-born scientist, so RPSC likes him.
- India's four science Nobel laureates by birth - C.V. Raman (Physics 1930), Har Gobind Khorana (Medicine 1968), S. Chandrasekhar (Physics 1983), Venkatraman Ramakrishnan (Chemistry 2009).
- Colour revolutions - Green for foodgrains, White for milk, Blue for fish and aquaculture, Yellow for oilseeds, Golden for horticulture and honey, Silver for eggs and poultry, Pink for onion and prawn, Grey for fertiliser.
- The pair RPSC plants as wrong is 'Blue Revolution - oilseeds'. Blue is FISH; oilseeds are YELLOW.
- Other awards to recognise - the Ramanujan Prize for young mathematicians and the Infosys Prize; in the state, science popularisation runs through Regional Science Centres and science parks under the Rajasthan Council of Science and Technology.

TRAP

The most-missed scientist fact: Har Gobind Khorana's 1968 Nobel was in PHYSIOLOGY OR MEDICINE, not Chemistry. RPSC has used that exact substitution. Two more in the same family - Chandrasekhar's prize was Physics 1983 and Ramakrishnan's Chemistry 2009, and Raman's 1930 prize was for the SCATTERING of light, not for spectroscopy or fluorescence.

REVISION RECAP

1. Seventeen essential nutrients; NPK primary, Ca-Mg-S secondary macronutrients, and eight micronutrients. Calcium is NOT a micronutrient.
2. N - chlorosis of older leaves; P - purplish tinge; K - scorched leaf margins; Zn - khaira of rice; Fe - chlorosis of young leaves.

3. Soil Health Card launched 19 February 2015 from Suratgarh, Sri Ganganagar; neem-coated urea mandatory the same year.
4. Seed chain - breeder (golden yellow), foundation (white), certified (azure blue); truthfully labelled is opal green.
5. Rajasthan leads India in bajra, rapeseed-mustard, guar, coriander, cumin, fenugreek, isabgol, henna and moth bean, plus wool and camels.
6. HHB-67 Improved - bajra, early maturing, downy-mildew resistant. RGC-936 and RGC-1003 - guar. Pusa Bold and Bio-902 - mustard. Mahi Kanchan, Mahi Dhawal and Pratap Makka - maize.
7. Guar is a kharif crop; barley is rabi. The mustard directorate (DRMR) is at Sewar, Bharatpur.
8. Dryland farming - below about 750 mm rain; the aim is moisture conservation, and khadin, mulching and short-duration varieties are the tools.
9. Water-use efficiency - flood 30-50 per cent, sprinkler 70-80 per cent, drip about 90 per cent. PMKSY 2015 = Har Khet Ko Pani + Per Drop More Crop.
10. Desert locust = Schistocerca gregaria; Locust Warning Organisation at Jodhpur under DPPQS Faridabad; Scheduled Desert Area in Rajasthan and Gujarat.
11. IPM keeps pests below the Economic Threshold Level, not at zero; Trichogramma is the egg parasitoid. Kinnow - Sri Ganganagar; orange - Jhalawar; olive - Lunkaransar and Bassi; date palm - Jaisalmer and Bikaner; pomegranate - Barmer and Jalore.
12. Sojat (Pali) is the henna capital with a GI tag; isabgol is Plantago ovata, ashwagandha Withania somnifera, safed musli Chlorophytum borivilianum, guggul Commiphora wightii.
13. Khejri is Prosopis cineraria, the state tree and desert agroforestry backbone; Rohida is Tecomella undulata, the state flower; Prosopis juliflora is the invasive vilayati babool.
14. National Forest Policy 1988 - 33 per cent target, ecology over revenue; JFM circular 1990; Van Dhan Kendras run by TRIFED.
15. Cattle - Rathi (best milch), Tharparkar, Kankrej, Nagauri. Sheep - Chokla (finest wool), Magra (carpet wool), Marwari, Nali, Sonadi. Goat - Sirohi (meat), Jakhrana of Alwar (milk).
16. Camel - state animal (domestic) 2014, state Act 2015; breeds Bikaneri, Jaisalmeri, Mewari, Kachchhi.
17. FMD, PPR and lumpy skin disease are viral; brucellosis is bacterial and zoonotic. NADCP 2019 targets FMD and brucellosis; Rashtriya Gokul Mission 2014 is for indigenous bovines.
18. Institutes - CAZRI Jodhpur 1959, CSWRI Avikanagar (Tonk) 1962, Udaipur Solar Observatory 1975, NRC Camel Bikaner 1984, CIAH Bikaner 1993, NRC Seed Spices Ajmer 2000; CEERI Pilani is the only CSIR lab in the state.

19. CSIR 1942, DST 1971, STIP 2020 the fifth policy, ANRF Act 2023 with the PM as ex-officio President.
20. Science days - 28 February, 11 May, 23 August, 22 December. Khorana's Nobel was in Medicine; Ramakrishnan's in Chemistry for the ribosome; Kothari of Udaipur chaired the 1964-66 Education Commission.